A Reliability-Constrained Time Delay Estimation Method for Multi-Sensor Array Systems

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***Abstract*—In strongly reverberant and multi-source sound environments, conventional generalized cross-correlation-based time delay estimation methods often suffer from degraded accuracy and stability due to multipath propagation and spurious correlations. This paper proposes a reliability-constrained TDE approach that integrates phase consistency and magnitude-squared coherence to evaluate frequency-domain reliability. By constructing a PC-MSC adaptive weighting scheme, the method emphasizes direct-path-dominant frequency components while suppressing unreliable bins, and further optimizes the weighting balance using a peak-to-sidelobe ratio criterion. To improve robustness in ambiguous multi-source scenarios, a multi-peak candidate selection strategy is introduced to retain significant correlation peaks instead of relying on a single maximum. Moreover, for near-field peak overlap caused by closely spaced sources, a structured multi-source decomposition framework incorporating inter-frame temporal consistency and near-field geometric constraints is developed to recover latent source-specific delays via soft assignment and alternating optimization. Simulation results in a realistic indoor room and real-world experiments demonstrate that the proposed method can improve localization accuracy and robustness under the evaluated low-SNR and moderate-to-strong reverberation conditions.**

Keywords—Sound source localization, time delay estimation, phase consistency, magnitude-squared coherence.

I. Introduction

Sound source localization (SSL) using microphone arrays is widely used in speech interaction, acoustic scene analysis, and intelligent monitoring [1]. In indoor environments, reverberation, background noise, and concurrent sources distort spatial cues, making robust time delay estimation (TDE) challenging.

Under such adverse sound conditions, a key challenge is how to achieve reliable TDE when conventional correlation-based methods suffer from spurious peaks, peak overlap, and instability under low signal-to-noise ratio and strong multipath propagation. Generalized cross-correlation GCC and its variants, such as GCC-PHAT, have been extensively studied due to their conceptual simplicity and low computational complexity [2]. To enhance performance in complex environments, previous works have proposed phase-based weighting strategies, structured signal modeling techniques, and advanced statistical formulations [3]-[5]. In addition, joint frameworks integrating SSL with speech enhancement, dereverberation, or source separation have been investigated to improve robustness against noise and reverberation [6]-[9]. While these methods can partially alleviate spatial aliasing and multipath effects, TDE performance remains highly sensitive to unreliable frequency components and ambiguous correlation peaks. In particular, most existing approaches rely on single-peak selection strategies, which are prone to pseudo-peak interference and peak broadening in near-field or closely spaced multi-source scenarios. As a result, the discrimination of multiple sources and the stability of delay estimation are still limited in practice [10].

To address these limitations, this paper proposes a reliability-constrained time delay estimation method based on phase consistency and magnitude-squared coherence features. The main contributions of this work are summarized as follows:

- A joint PC–MSC reliability-aware weighting strategy is introduced to emphasize relatively reliable frequency components in GCC-based TDE;
- A multi-peak candidate selection strategy is introduced to preserve significant correlation peaks as observational inputs, improving TDE stability in ambiguous multi-source environments;
- A structured multi-source decomposition model with temporal and near-field geometric constraints is proposed for overlapping TDOA peaks.

II. Fundamentals of Time Delay Estimation and Feature Modeling

A. Microphone Array Signal Model

Consider a microphone array consisting of m microphones deployed on a two-dimensional plane, with their spatial position vectors denoted as p_m. Assume that a sound source is located at position $s_k = (x_k, y_k)$ during the t-th time frame. The signal received by the m microphone can be expressed as

$$x_m(t,k) = s(t-\tau_{m,k}) + v_m(t,k), \tag{1}$$

where $s(t)$ denotes the source signal, $\tau_{m,k}$ is the propagation delay from the source to the m-th microphone, $c$represents the speed of sound, and $v_m(t,k)$ denotes the interference term,

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which includes ambient noise and reverberation components.

B. *Time Delay Estimation Based on Generalized Cross-Correlation*

For an arbitrary microphone pair (i,j), let $X_i(f,t)$ and $X_j(f,t)$ denote the short-time Fourier transforms (STFTs) of the received signals at frequency bin f and frame t. The corresponding cross-power spectral density is defined as

$$G_{ij}(f,t) = X_i(f,t)X_j^*(f,t), \tag{2}$$

where $(\cdot)^*$ denotes complex conjugation.

Within the generalized cross-correlation framework, an appropriate frequency-domain weighting function is applied to the cross-spectrum, followed by an inverse Fourier transform to obtain the time-domain delay correlation function:

$$r_{ij}(\tau,t) = \mathcal{F}^{-1}\{W(f,t)G_{ij}(f,t)\}, \tag{3}$$

where $\mathcal{F}^{1}\cdot$ denotes the inverse Fourier transform and $W(f,t)$ represents the frequency-domain weighting function. A widely used variant is GCC-PHAT, which adopts the phase transform (PHAT) weighting:

$$\tilde{G}_{ij}(f) = \frac{G_{ij}(f,t)}{|G_{ij}(f,t)|+\epsilon}, \tag{4}$$

where ϵ is a small constant introduced for numerical stability. While GCC-PHAT effectively preserves phase information, its performance degrades significantly under strong reverberation and low signal-to-noise ratio (SNR) conditions.

C. *Phase Consistency Feature Modeling*

To quantify the temporal stability of frequency-domain phase differences, the PC feature is introduced. For a given microphone pair (i,j), the PC over a window of N frames at frequency f is defined as

$$\mathrm{PC}_{ij}(f,t) = \frac{1}{N}\left|\sum_{k=1}^{N} e^{j[\angle X_i(f,t) - \angle X_j(f,t)]}\right|, \tag{5}$$

where $\angle(\cdot)$ denotes the phase operator and k is the frame index.

D. *Magnitude-Squared Coherence Feature Modeling*

In addition to phase stability, the MSC is introduced as a second frequency-domain feature to characterize the linear correlation strength between the magnitudes of two microphone signals. For a microphone pair (i,j), the MSC at frequency bin f and frame t is defined as

$$\mathrm{MSC}_{ij}(f,t) = \frac{\left|E\left[X_i(f,t)X_j^*(f,t)\right]\right|^2}{E\left[|X_i(f,t)|^2\right]E\left[|X_j(f,t)|^2\right]}. \tag{6}$$

where $E(\cdot)$ denotes mathematical expectation. In practical implementations, the expectation operator is approximated by averaging over multiple short-time Fourier transform frames.

III. PC-MSC-Based Robust TDE and Multi-Source Decomposition

In this work, PC and MSC are used as heuristic frequency-domain reliability indicators rather than strict sufficient conditions for direct-path dominance. Based on this motivation, they are combined to construct an adaptive weighting strategy for GCC-based delay estimation. Fig. 1 illustrates the structure of the proposed PC–MSC reliability-constrained time delay estimation method (PCMSC-GCC).

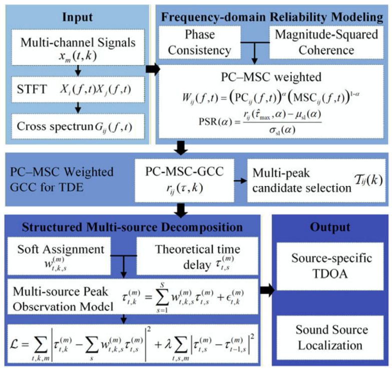

Fig. 1. Structure of PC–MSC reliability-constrained time delay estimation method (PCMSC-GCC)

A. *PC–MSC-Based Frequency-Domain Adaptive Weighted GCC for Time Delay Estimation*

PC and MSC are treated as complementary reliability cues in the frequency domain. Specifically, PC reflects temporal phase stability, while MSC characterizes inter-channel spectral coherence. Based on these two cues, the adaptive weighting function is defined as

$$W_{ij}(f,t) = \left(\mathrm{PC}_{ij}(f,t)\right)^{\alpha}\left(\mathrm{MSC}_{ij}(f,t)\right)^{1-\alpha}, \quad \alpha \in [0,1] \tag{7}$$

where α is a weighting control parameter that balances the contributions of phase stability and amplitude correlation.

To determine an appropriate value of α, the peak-to-sidelobe ratio (PSR) is employed as an optimization criterion to maximize the separability of the correlation peaks:

$$\mathrm{PSR}(\alpha) = \frac{r_{ij}(\hat{\tau}_{\max},\alpha) - \mu_{\mathrm{sl}}(\alpha)}{\sigma_{\mathrm{sl}}(\alpha)}, \tag{8}$$

where $\hat{\tau}_{\max}$ denotes the location of the main correlation peak, μ_{sl}, σ_{sl} represent the mean and standard deviation of the sidelobe region, respectively, excluding a neighborhood around the main peak. The optimal weighting parameter is then defined as

$$\alpha^{\star} = \arg\max_{\alpha\in[0,1]} \mathrm{PSR}(\alpha), \tag{9}$$

Accordingly, the PC–MSC weighted cross-correlation (PCC) function is obtained as

$$r_{ij}(\tau,k) = \mathcal{F}^{-1}\left\{W_{ij}(f,t)\tilde{G}_{ij}(f,t)\right\}, \tag{10}$$

Under strong reverberation conditions, $r_{ij}(\tau,k)$ often exhibits a multi-peak structure, where selecting only the global maximum is prone to pseudo-peak interference. Therefore, the top Q most significant local maxima are retained to form a set of candidate time delay estimates:

$$\mathcal{T}_{ij}(k) = \left\{\tau_{t,k}^{(m)}\right\}_{q=1}^{Q}, \tag{11}$$

which provides a robust observation basis for subsequent multi-source decomposition.

B. Structured Decomposition of Multi-Source Time Delay Candidate Peaks

In near-field multi-source scenarios, closely spaced sound sources and nonlinear propagation effects result in highly similar time delays at the same microphone pair. Consequently, correlation peaks corresponding to different sources tend to overlap and exhibit mixed contributions, making conventional single-peak selection strategies insufficient.

To address this issue, the extracted time delay candidate peaks are modeled as mixed observations generated by multiple sound sources. By incorporating inter-frame temporal consistency and near-field geometric constraints, a structured decomposition framework is proposed to recover the latent time delays of individual sources while suppressing spurious peaks.

Let $\tau_{t,k}^{(m)}$ denote the k time delay candidate extracted from microphone pair (i,j) at frame n. This candidate may contain contributions from multiple sound sources and is modeled as

$$\tau_{t,k}^{(m)}=\sum_{s=1}^{S} w_{t,k,s}^{(m)}\tau_{t,s}^{(m)}+\epsilon_{t,k}^{(m)}, \tag{12}$$

where S is the number of sound sources, $\tau_{t,s}^{(m)}$ represents the true time delay of source $s$at microphone pair, $w_{t,k,s}^{(m)}$ represents the soft assignment weight corresponding to the candidate peak contributed by the s sound source, the soft assignment weights are defined by a normalized exponential function in (12), ensuring $\sum_{s} w_{t,k,s}^{(m)}=1$, and $\epsilon_{t,k}^{(m)}$ accounts for noise and residual interference.

Sound source positions evolve smoothly over time, and their corresponding near-field time delays exhibit temporal continuity. Based on the estimated delays from the previous frame, a reference delay is predicted for the current frame. To avoid unstable hard assignments under peak overlap, the soft assignment weights are defined as

$$w_{t,k,s}^{(m)}=\exp\left(-\frac{\left|\tau_{t,k}^{(m)}-\tau_{t-1,s}^{(m)}\right|^{2}}{\sigma_{\tau}^{2}}\right)\Bigg/\sum_{s'=1}^{S}\exp\left(-\frac{\left|\tau_{t,k}^{(m)}-\tau_{t-1,s'}^{(m)}\right|^{2}}{\sigma_{\tau}^{2}}\right), \tag{13}$$

where $\tau_{t-1,s}^{(m)}$ denotes the estimated delay of source S in the previous frame, and σ_{τ} controls the tolerance of delay matching.

C. Near-Field Geometric Consistency Constraint

To further enhance the physical plausibility of the estimated time delays, a near-field sound geometric constraint is incorporated. For a sound source located at position $\boldsymbol{x}_s(t)$, the theoretical time delay at microphone pair m is given by

$$\tau_{t,s}^{(m)}=\frac{1}{c}\left(\left\|\boldsymbol{x}_s(t)-\boldsymbol{r}_i\right\|-\left\|\boldsymbol{x}_s(t)-\boldsymbol{r}_j\right\|\right), \tag{14}$$

where microphone pair m corresponds to microphones (i,j) .respectively, and c is the speed of sound. By introducing this geometric relationship, the time delay decomposition problem is further constrained by spatial consistency, linking the estimated delays across different microphone pairs through a common source position.

D. Joint Optimization and Time Delay Estimation

Combining the above modeling components, the estimation of source-specific time delays is formulated as the minimization of the following objective function:

$$\mathcal{L}=\sum_{t,k,m}\left|\tau_{t,k}^{(m)}-\sum_{s} w_{t,k,s}^{(m)}\tau_{t,s}^{(m)}\right|^{2}+\lambda\sum_{t,s,m}\left|\tau_{t,s}^{(m)}-\tau_{t-1,s}^{(m)}\right|^{2}. \tag{15}$$

where the first term enforces accurate reconstruction of the observed candidate peaks, the second term imposes inter-frame temporal consistency, and λ is a regularization parameter that balances the two terms. An alternating iterative optimization strategy is adopted, in which the soft assignment weights and the source-specific time delays are updated iteratively until convergence.

IV. Experimental and Results Analysis

To systematically evaluate the effectiveness and robustness of the proposed PC-MSC reliability-constrained time delay estimation and sound source localization method under complex sound conditions, simulation and real-world indoor experiments are conducted.

A. Simulation Experiments

Simulations were conducted in a conference room using image-source room impulse responses with an average wall reflection coefficient of 0.62. An eight-element circular microphone array with radius 0.045 m was placed at (3.75, 3.5, 1.0) m. Speech signals with additive white Gaussian noise were used to evaluate different SNR and RT_{60} conditions.

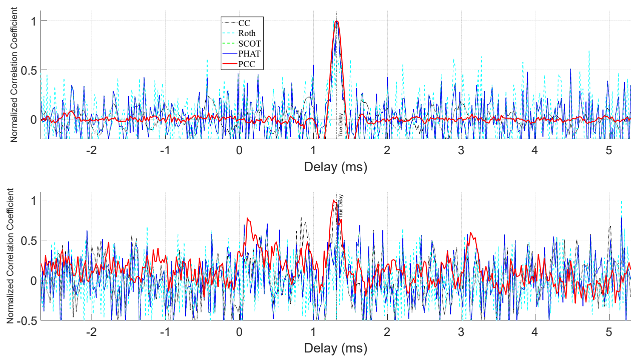

Fig. 2. Normalized cross-correlation function obtained with PCC weighting

Fig. 2 shows the normalized cross-correlation function obtained with PCC weighting, it can be observed that, compared with phase-normalization-based algorithms, although the correlation function still exhibits certain sidelobes under strong reverberation and noisy conditions, the main peak remains clearly distinguishable. Moreover, the correlation peak width is relatively concentrated, and the amplitude of the main peak is significantly higher than that of adjacent spurious peaks.

Localization performance is quantified using root mean square error (RMSE) of source position estimates, with comparisons against four experiment algorithms across varying noise and reverberation levels.

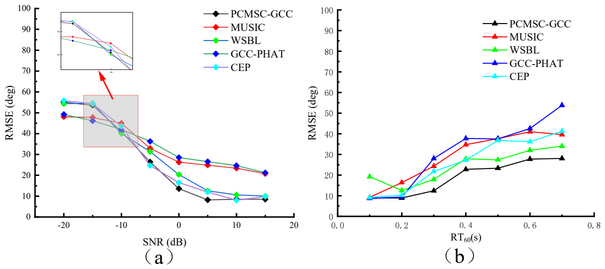

Fig. 3. Localization RMSE comparison of five algorithms under different noise and reverberation conditions. (a) RMSE versus SNR with fixed RT_{60}=0.3 s. (b) RMSE versus RT_{60} with fixed SNR=-5 dB

Fig. 3 (a) illustrates the localization RMSE as a function of SNR under a fixed RT_{60}=0.3 s. As the SNR increases, the RMSE of all five algorithms decreases. However, the proposed PCMSC-GCC method consistently achieves the lowest RMSE across the entire SNR range.

Fig. 3 (b) illustrates the localization RMSE as a function of RT_{60} under a fixed SNR=-5 dB. As RT_{60} increases, spectral coherence is significantly degraded and multipath-induced spurious peaks become more pronounced, leading to a general increase in RMSE for all methods. Nevertheless, in the evaluated moderate-reverberation range, PCMSC-GCC achieves lower RMSE than the compared methods in most tested cases, indicating improved robustness to early reflections and moderate reverberation.

TABLE I Ablation study under SNR = -5 dB and RT_{60} = 0.3 s

Method variant	Ablation study under SNR = -5 dB and RT_{60} = 0.3 s
	RMSE / m
GCCPHAT	0.3477 ± 0.0165
GCC + PC	0.3416 ± 0.0153
GCC + MSC	0.3363 ± 0.0153
PCMSC-GCC with PSR	0.3290 ± 0.0115

TABLE I is ablation study under SNR = -5 dB and RT_{60} = 0.3 s. The results show that PC and MSC weighting both improve the GCC-PHAT baseline, while the PSR-based PCMSC-GCC achieves the best performance.

B. *Real-World Experimental Evaluation*

To assess the feasibility of the proposed method under real sound conditions and hardware nonidealities, indoor experiments were conducted in a room matching the simulation dimensions.

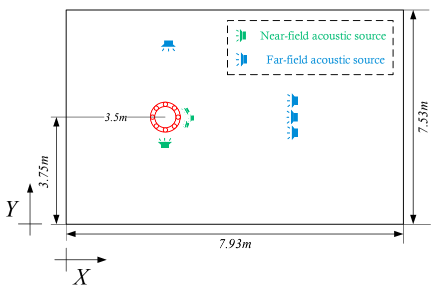

Fig. 4. Indoor microphone array and sound source configuration

Fig. 4 is indoor microphone array and sound source configuration, with measured $RT_{60} \approx$ 743 ms and background noise ≈ 23 dBA. Data acquisition utilized Orange Pi Zero 2 and Radxa RK3399 multichannel boards. The eight-element circular microphone array (radius 0.045 m) was identical to the simulated configuration. Near- and far-field source positions were evaluated using broadband speech excitations, with each measurement repeated 30 times and averaged to minimize random variability.

TABLE II. Near-field sound source field measurement

Algorithm	Near-field sound source field measurement		
	Near-Field Position	True	Error
PCMSC-GCC	(4.13, 3.32) (4.00, 3.55) (4.14, 3.64) (3.90, 3.24)	(4.05,3.2) (4.05,3.5) (4.05,3.8) (3.75,3.2)	0.1345 m
MUSCI	(3.78, 3.07) (3.99, 3.67) (4.15, 3.60) (3.96, 3.28)		0.23 m
WSBL	(4.16, 3.42) (4.01, 3.35) (3.78, 3.92) (3.58, 3.24)		0.2175

TABLE III. Far-field sound source field measurement

Algorithm	Far-field sound source field measurement		
	Far-Field Position	True	Error
PCMSC-GCC	(5.66, 4.17) (5.52, 4.44) (5.66, 2.83) (2.81, 5.27)	(5.75,3.2) (5.75,3.5) (5.75,3.8) (3.75,5.5)	0.97m
MUSCI	(5.19, 2.08) (5.37, 4.68) (5.54, 2.57) (2.57, 5.12)		1.24 m
WSBL	(5.56, 4.40) (5.48, 2.50) (5.09, 5.02) (4.84, 5.18)		1.19 m

TABLE II summarizes near-field sound source field measurement data and TABLE III summarizes Far-field sound source field measurement data. In near-field scenarios, the proposed PCMSC-GCC achieves an average error of 0.1345 m, reducing errors by ~41.5% and ~38.2% relative to MUSIC and WSBL, respectively, by effectively mitigating TDOA peak overlap and spurious peak influence under strong multipath. In far-field scenarios, it yields 0.97 m average error, outperforming MUSIC and WSBL by ~21.8% and ~18.5%. Far-field errors are higher due to amplified angular errors in coordinate reconstruction and degraded coherence from reverberation and occlusion, yet the proposed method maintains superior robustness.

V. CONCLUSION

This paper proposes a reliability-constrained time delay estimation method integrating phase consistency and magnitude-squared coherence features for strongly reverberant environments with multiple concurrent sources. Frequency-domain adaptive weighting enhances direct-path components while suppressing unreliable bins, improving accuracy and stability over conventional GCC under low SNR and multipath interference. A structured decomposition framework incorporating cross-frame consistency and near-field geometric constraints, with soft assignment and joint optimization, resolves near-field peak overlap by recovering individual source delays and suppressing spurious peaks. Simulation and real-world experiments show lower RMSE than the compared methods under the evaluated moderate-to-strong reverberation and low-SNR conditions, supporting the effectiveness of the proposed method in the tested scenarios.

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